

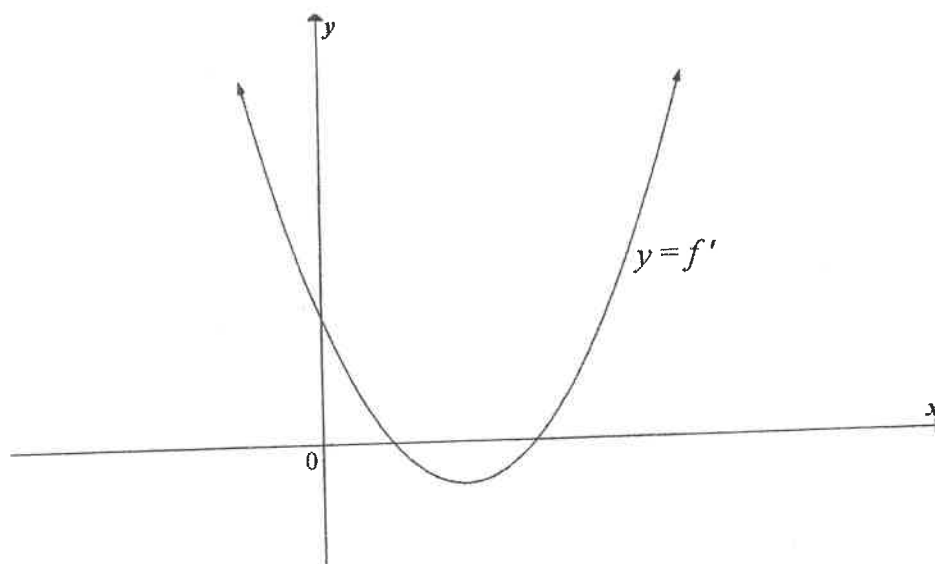
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Calculus II Paper 1

WORKING BACKWARDS FROM DERIVATIVE

QUESTION 1

The graph of $y = ax^2 + bx + c$ below represents the derivative of f .
It is given that $f'(1) = 0$, $f'(3) = 0$ and $f'(0) = 6$.



- 1.1 Write down the x -coordinates of the stationary points of f . (2)
- 1.2 For which value(s) of x is f strictly decreasing? (2)
- 1.3 Explain at which value of x the stationary point of f will be a local minimum. (2)
- 1.4 Determine the x -coordinate of the point of inflection of f . (1)
- 1.5 For which value(s) of x is f concave up? (2)

[9]

RATES OF CHANGE

QUESTION 2

The mass of a baby in the first 30 days of life is given by

$$M(t) = t^3 - 9t^2 + 3\,000 \quad ; \quad 0 \leq t \leq 30.$$

t is the time in days and M is the mass of the baby in grams.

- 2.1 Write down the mass of the baby at birth. (1)
- 2.2 A baby's mass usually decreases in the first few days after birth.
On which day will the baby's mass return to its birth mass? (4)
- 2.3 On which day will this baby have a minimum mass? (4)
- 2.4 On which day will the baby's mass be decreasing the fastest? (2)

[11]

QUESTION 3

A scientist adds a bactericide into a culture of bacteria. The number of bacteria present t hours after the bactericide was introduced is given by the formula:

$$B(t) = 1000 + 50t - 5t^2 \text{ where } B(t) \text{ is the number of bacteria, in millions.}$$

- (a) How many bacteria were in the culture when the bactericide was introduced?
- (b) Calculate the rate of change of the number of bacteria with respect to time three hours after the bactericide was added.
- (c) When does the population of bacteria start to decrease?
- (d) When will the whole culture of bacteria be exterminated?

QUESTION 4

A stone is thrown vertically upwards and its height (in metres) above the ground at time t (in seconds) is given by:

$$h(t) = 35 - 5t^2 + 30t$$

- (a) Determine its initial height above the ground.
- (b) Determine the initial speed with which it was thrown.
- (c) Determine the maximum height above the ground that the stone reached.
- (d) How fast was the stone travelling when it reached a height of 60 metres above the ground on the way down?
- (e) How fast was the stone travelling when it hit the ground?

QUESTION 5

A tourist travels in a car over a mountainous pass during his trip. The height above sea level of the car, after t minutes, is given as $s(t) = 5t^3 - 65t^2 + 200t + 100$ metres. The journey lasts 8 minutes.

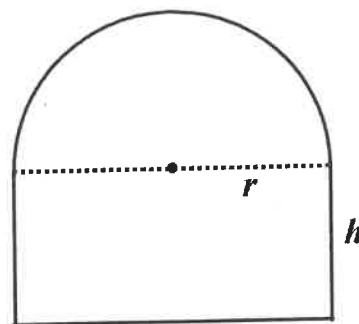
- 5.1 How high is the car above sea level when it starts its journey on the mountainous pass? (2)
- 5.2 Calculate the car's rate of change of height above sea level with respect to time, 4 minutes after starting the journey on the mountainous pass. (3)
- 5.3 Interpret your answer to QUESTION 5.2. (2)
- 5.4 How many minutes after the journey has started will the rate of change of height with respect to time be a minimum? (3)

MINIMUM - / MAXIMUM PROBLEMS

QUESTION 6

A builder wishes to construct a steel window frame in the shape of a rectangle with a semi-circular part on top. The radius of the semi-circular part is r metres and the width of the rectangular part is h metres.

- (a) Write down, in terms of h and r
- (1) the steel perimeter (P) of the frame.
 - (2) the area enclosed by the frame.
- (b) The area enclosed by the frame is to be 4 square metres.

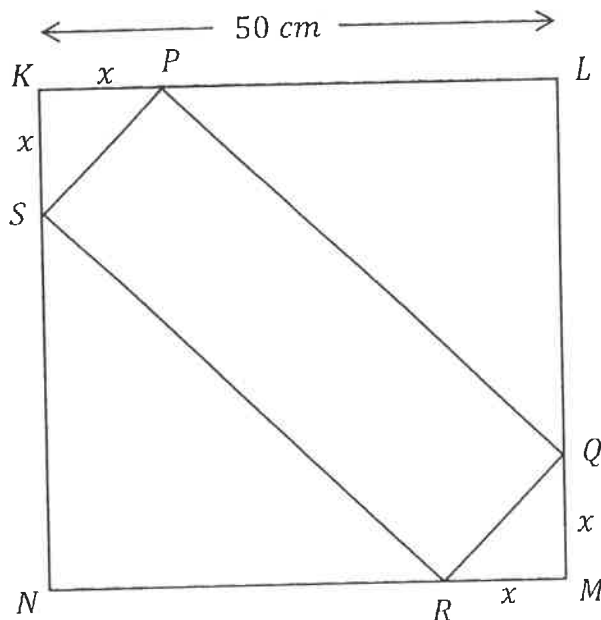


Show that the perimeter (P) is $P = \left(\frac{\pi}{2} + 2\right)r + \frac{4}{r}$

- (c) If the steel for the frame costs R10 per metre, calculate the value of r for which the total cost of the steel will be a minimum.

QUESTION 7

$KLMN$ is a square in the figure below. $KL = 50$ cm. Rectangle $PQRS$ is drawn inside square $KLMN$ so that $KP = KS = RM = QM = x$.

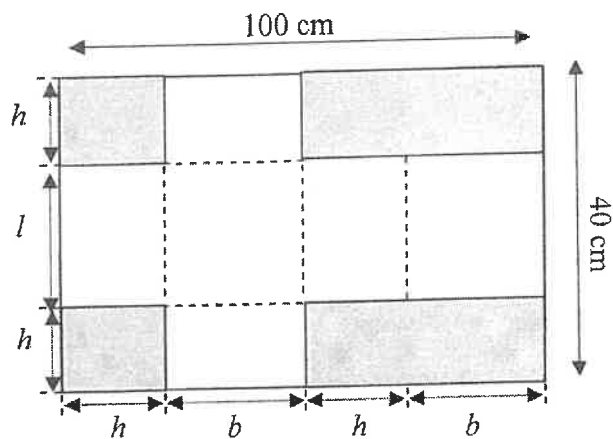
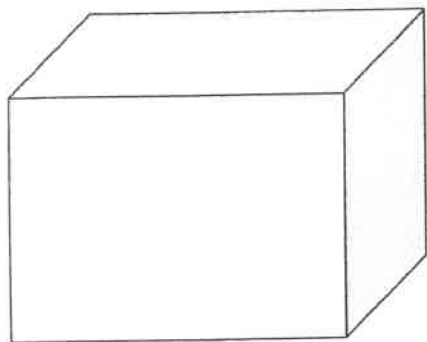


- 7.1 Show that the area (A) of rectangle $PQRS$ can be expressed as:

$$A = 100x - 2x^2 \quad (4)$$

- 7.2 Hence determine the maximum area of $PQRS$. Show all calculations. (3)

QUESTION 8

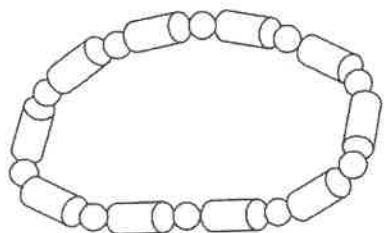


A box is made from a rectangular piece of cardboard, 100 cm by 40 cm, by cutting out the shaded areas and folding along the dotted lines as shown in the diagram above.

- 8.1 Express the length l in terms of the height h . (1)
- 8.2 Hence prove that the volume of the box is given by $V = h(50 - h)(40 - 2h)$ (3)
- 8.3 For which value of h will the volume of the box be a maximum? (5)

QUESTION 9

A necklace is made by using 10 wooden spheres and 10 wooden cylinders. The radii, r , of the spheres and the cylinders are exactly the same. The height of each cylinder is h . The wooden spheres and cylinders are to be painted. (Ignore the holes in the spheres and cylinders.)

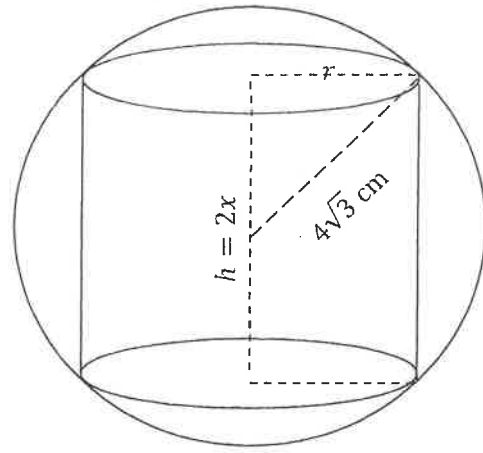


$V = \pi r^2 h$	$S = 2\pi r^2 + 2\pi r h$
$V = \frac{4}{3}\pi r^3$	$S = 4\pi r^2$

- 9.1 If the volume of a cylinder is 6 cm^3 , write h in terms of r . (1)
- 9.2 Show that the total surface area (S) of all the painted surfaces of the necklace is equal to $S = 60\pi r^2 + \frac{120}{r}$ (4)
- 9.3 Determine the value of r so that the least amount of paint will be used. (4)

QUESTION 10

A cylinder fits neatly into a sphere with radius of $4\sqrt{3}$ cm



$$V(\text{cylinder}) = \pi r^2 h$$

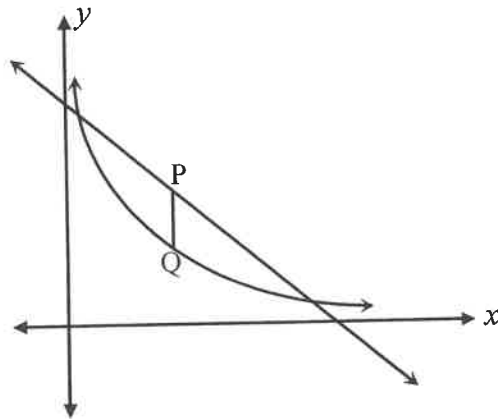
10.1 If the height of the cylinder is $2x$, show that the radius of the cylinder is given by $r = \sqrt{48 - x^2}$ cm (2)

10.2 Hence, show that the volume of the cylinder in terms of x is given by $V = 96\pi x - 2\pi x^3$ (3)

10.3 Calculate the height of the cylinder so that the volume is a maximum. (4)

QUESTION 11

The line with equation $x + y = 4$ and the graph of the function $xy = 1$ for $x > 0$ are represented in the diagram alongside. Line segment PQ varies in length between the two points of intersection of the graphs. Determine the maximum length of PQ.

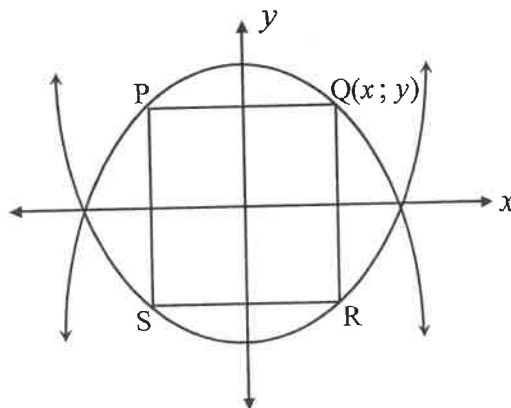


QUESTION 12

PQRS is a rectangle with sides parallel to the axes. The points P, Q, R and S are points on the parabolas

$$y = -\frac{1}{3}x^2 + 4 \quad \text{and} \quad y = \frac{1}{6}x^2 - 2.$$

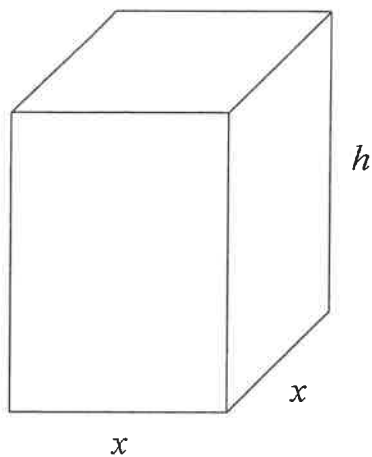
- Express PQ and QR in terms of x .
- Calculate the maximum area of PQRS.



QUESTION 13

The sketch shows a box with a square base. The volume of the box has to be 125 cm^3 .

Let the length of the base be x and the height of the box h :



Determine the minimum surface area of the box.

[6]

Calculus 2: SolutionsWorking from the derivativeQuestion 1:

1.1. $\underline{x=1}$, or $\underline{x=3}$,

1.2. $1 < \underline{x} < 3$,

1.3. $x=3$ f decreasing before $x=3$ and increasing after $x=3$

1.4. $\underline{x=2}$,

1.5. $\underline{x > 2}$,

Rates of changeQuestion 2:

2.1. $M(0) = \underline{3000 \text{ grams}}$,

2.2. $t^3 - 9t^2 + 3000 = 3000$

$t^3 - 9t^2 = 0$

$t^2(t - 9) = 0$

$\underline{t=0}$, ; $\underline{t=9}$,

 $\therefore$ on the ninth day.

$$2.3. \quad M'(t) = 3t^2 - 18t = 0$$

$$+ (3t - 18) = 0$$

$$t = 0 \quad \text{OR} \quad \underline{t = 6}$$

$$M(6) = (6)^3 - 9(6)^2 + 3000$$

$$= \underline{2892 \text{ g}}$$

$\therefore$ On day 6,

$$2.4. \quad M''(t) = 6t - 18 = 0$$

$$6t = 18$$

$$\underline{t = 3}$$

$\therefore$ On day 3

Question 3

a) $B(0) = \underline{1000},$

b) $B'(t) = 50 - 10t$

$$B'(3) = 50 - 10(3)$$

$$= \underline{20},$$

$\therefore$ Increase at a rate of 20 per hour

c) $B'(t) = 0$

$$50 - 10t = 0$$

$$\underline{t = 5},$$

$\therefore$ After 5 hours

$$\begin{aligned}
 d) \quad 1000 + 50t - 5t^2 &= 0 \\
 t^2 - 10t - 200 &= 0 \\
 (t-20)(t+10) &= 0 \\
 \underline{t=20}, & \quad \text{or} \quad t=-10 \\
 & \quad \text{N.A.}
 \end{aligned}$$

$\therefore$ After 20 hours

Question 4:

$$a) h(0) = \underline{35 \text{ m}}$$

$$\begin{aligned}
 b) \quad v(t) &= h'(t) = -10t + 30 \\
 v(0) &= 30 \text{ m/s}
 \end{aligned}$$

$$\begin{aligned}
 c) \quad v(t) &= 0 \quad \therefore -10t + 30 = 0 \\
 & \quad \underline{t = 3 \text{ s}},
 \end{aligned}$$

$$\begin{aligned}
 h(3) &= 35 - 5(3)^2 + 30(3) \\
 &= \underline{80 \text{ m}}
 \end{aligned}$$

$$\begin{aligned}
 d) \quad h(t) &= 35 - 5t^2 + 30t = 0 \\
 -5t^2 + 30t - 25 &= 0 \\
 t^2 - 6t + 5 &= 0 \\
 (t-1)(t-5) &= 0 \\
 \underline{t=1 \text{ s}} \quad \text{or} \quad \underline{t=5 \text{ s}}
 \end{aligned}$$

$$\begin{aligned}
 v(5) &= -10(5) + 30 \\
 &= \underline{-20 \text{ m/s}}
 \end{aligned}$$

c)

$$h(t) = 0$$

$$-5t^2 + 30t + 35 = 0$$

$$t^2 - 6t - 7 = 0$$

$$(t - 7)(t + 1) = 0$$

$$\underline{t = 7},$$

$$\text{OR } t = -1$$

N.A

$$\begin{aligned} V(7) &= -10(7) + 30 \\ &= \underline{-40 \text{ m/s}} \end{aligned}$$

Question 5:

$$5.1. \quad s(0) = 100 \text{ m}$$

$$\begin{aligned} 5.2. \quad s'(t) &= 15t^2 - 130t + 200 \\ s'(4) &= 15(4)^2 - 130(4) + 200 \\ &= \underline{-80 \text{ m/s}} \end{aligned}$$

$$5.4. \quad s''(t) = 30t - 130$$

$$\therefore 30t - 130 = 0$$

$$t = \frac{13}{3} \text{ s}$$

$$= \underline{4.3 \text{ min}}$$

5.3. His height above sea level is decreasing at a rate of 80 m/s

Maximum/Minimum ProblemsQuestion 6

$$a. 1) \quad P = \pi r + 2h + 2r$$

$$2) \quad A = \frac{1}{2} \pi r^2 + 2\pi h$$

b)

$$\frac{1}{2} \pi r^2 + 2\pi h = 4$$

$$2\pi h = 4 - \frac{1}{2} \pi r^2$$

$$h = \frac{4}{2\pi} - \frac{\pi r^2}{4\pi}$$

$$h = \frac{2}{\pi} - \frac{\pi r}{4}$$

into P

$$P = \pi r + 2 \left(\frac{2}{\pi} - \frac{\pi r}{4} \right) + 2\pi$$

$$= \pi r + \frac{4}{\pi} - \frac{\pi r}{2} + 2\pi$$

$$= \frac{\pi r}{2} + 2\pi + \frac{4}{\pi}$$

$$= \left(\frac{\pi}{2} + 2 \right) r + \frac{4}{\pi}$$

$$c) \quad P' = \frac{\pi}{2} + 2 - 4r^{-2} = 0$$

$$\frac{4}{r^2} = \frac{\pi}{2} + 2$$

$$\frac{8}{r^2} = \pi + 4$$

$$r^2 = \frac{8}{\pi + 4}$$

$$r = \sqrt{\frac{8}{\pi + 4}}$$

Question 7

$$\begin{aligned}
 7.1 \quad PS^2 &= x^2 + x^2 && \text{pythag} \\
 PS^2 &= 2x^2 \\
 \underline{PS} &= \underline{\sqrt{2} \cdot x},
 \end{aligned}$$

$$SN = NR = 50 - x$$

$$\begin{aligned}
 SR^2 &= SN^2 + NR^2 && \text{pythag} \\
 &= (50-x)^2 + (50-x)^2 \\
 &= 2(50-x)^2
 \end{aligned}$$

$$\bullet \quad \underline{SR = \sqrt{2} (50-x)},$$

$$\begin{aligned}
 \text{Area} &= PS \cdot SR \\
 &= \sqrt{2} \cdot x \times \sqrt{2} (50-x) \\
 &= 2x (50-x) \\
 &= \underline{100x - 2x^2},
 \end{aligned}$$

$$\begin{aligned}
 7.2 \quad A' &= 100 - 4x = 0 \\
 x &= 25 \text{ cm}
 \end{aligned}$$

$$\begin{aligned}
 \therefore \text{max area} &= 100(25) - 2(25)^2 \\
 &= \underline{1250 \text{ cm}^2},
 \end{aligned}$$

Question 8

$$\begin{aligned}
 8.1 \quad 2h + l &= 40 \\
 \underline{l} &= \underline{40 - 2h},
 \end{aligned}$$

$$\begin{aligned}
 8.2 \quad 2b &= 100 - 2h \\
 \underline{b} &= \underline{50 - h},
 \end{aligned}$$

(7)

$$V = l \cdot b \cdot h$$

$$= (40-2h)(50-h) \cdot h,$$

$$8.3. \quad V = h(2000 - 140h + h^2)$$

$$= 2000h - 140h^2 + h^3$$

$$V' = 2000 - 280h + 3h^2 = 0$$

$$3h^2 - 280h + 2000 = 0$$

$$h = \frac{280 \pm \sqrt{(280)^2 - 4(3)(2000)}}{2(3)}$$

$$h = 85,54 \text{ cm} \quad \text{OR } \underline{h = 7,79 \text{ cm.}}$$

N.A

Question 9:

$$9.1. \quad V = \pi r^2 h = 6$$

$$h = \frac{6}{\pi r^2}$$

$$9.2. \quad S = 10 \left(2\pi r^2 + 2\pi r h \right) + 10 \left(4\pi r^2 \right)$$

$$= 20\pi r^2 + 20\pi r h + 40\pi r^2$$

$$= 60\pi r^2 + 20\pi r \left(\frac{6}{\pi r^2} \right)$$

$$= \underline{60\pi r^2 + \frac{120}{r}}$$

$$9.3. \quad S = 60\pi r^2 + 120r^{-1}$$

$$S' = 120\pi r - 120r^{-2} = 0$$

$$120\pi r^3 = 120$$

$$r^3 = \frac{1}{\pi}$$

$$r = \sqrt[3]{1/\pi}$$

$$\underline{r = 0,68 \text{ cm.}}$$

Question 10:

10.1. $(4\sqrt{3})^2 = x^2 + r^2$ pythag
 $48 = x^2 + r^2$
 $r^2 = 48 - x^2$
 $r = \sqrt{48 - x^2}$

10.2 $V = \pi r^2 h$
 $= \pi r^2 2x$
 $= \pi (\sqrt{48 - x^2})^2 2x$
 $= 2\pi x (48 - x^2)$
 $= 96\pi x - 2\pi x^3$

10.3. $V' = 96\pi - 6\pi x^2 = 0$
 $16 - x^2 = 0$
 $x = 4$

$L = 2(x) = 8 \text{ cm}$

Question 11:

$y = -x + 4$
 $y = \frac{1}{5x}$

$PQ = -x + 4 - \frac{1}{5x}$
 $= -x + 4 - x^{-1}$

$PQ' = -1 + x^{-2} = 0$

$\frac{1}{x^2} = 1$

$x^2 = 1$

$x = -1$

or

$x = 1$

N.A

$$\therefore PQ = - (1) + 4 - \frac{1}{1}$$

$$= \underline{2 \text{ units}}$$

Question 12

a) $PQ = 2x,$

$$QR = -\frac{1}{3}x^2 + 4 - \left(-\frac{1}{6}x^2 - 2\right)$$

$$= \underline{-\frac{1}{2}x^2 + 6}$$

b) Area = $PQ \cdot QR$

$$= 2x \left(-\frac{1}{2}x^2 + 6\right)$$

$$= -x^3 + 12x$$

$$A' = -3x^2 + 12 = 0$$

$$x^2 - 4 = 0$$

$$(x-2)(x+2) = 0$$

$$\underline{x=2}, \text{ or } x = -2$$

N.A

$$\text{Area} = - (2)^3 + 12(2)$$

$$= \underline{16 \text{ units}^2}$$

Question 13

$$V = x \cdot x \cdot h = 125$$

$$h = \frac{125}{x^2}$$

$$S.A = 2x^2 + 4xh \quad \text{sub } h = \frac{125}{x^2}$$

$$S = 2x^2 + 4x \left(\frac{125}{x^2} \right)$$

$$= 2x^2 + 500x^{-1}$$

$$S' = 4x - 500x^{-2} = 0$$

$$4x - \frac{500}{x^2} = 0$$

$$4x^3 - 500 = 0$$

$$4x^3 = 500$$

$$x^3 = 125$$

$$\underline{x = 5},$$

$$S = 2(5)^2 + \frac{500}{5}$$

$$= \underline{150 \text{ units}^2},$$